

Problem 1.1 (2 pts)

A flight test technique used to estimate the drag coefficient of early airships involved accelerating the vehicle up to its maximum forward speed, cutting the engines, and then measuring how the vehicle speed slowed over time. Use Euler integration to find the vehicle speed after it has coasted without engine power for 30 seconds. Use an integration time step of 0.1 seconds and assume an initial airspeed of 70 ft/s.

$$m \frac{dV}{dt} = -\frac{1}{2} \rho V^2 \nabla^{2/3} C_D \longrightarrow \left[\frac{dV}{dt} = -\frac{1}{2m} \rho V^2 \nabla^{2/3} C_D \text{ (EOM)} \right]$$

mass, $m = 200$ slugs

density, $\rho = 0.002377$ slug/ft³

airship volume, $\nabla = 89,300$ ft³

drag coefficient, $C_D = 0.044$

airship velocity after coasting 30 seconds = 33.3378 [ft/s]

Euler Integration: $X_{k+1} = X_k + h f[X_k]$

* $\frac{dx_k}{dt} = f[X_k] = 1$ if time
included in state vector

Simulation loop: 1) Initial state vector: $X_{old} = X_0$

2) Set control variables

3) Save (store X_{old})

4) Euler step: $X_{new} = X_{old} + h f[X_{old}]$

5) Reset: $X_{old} = X_{new}$

← EOM eqn

In this problem: • Initial state vector: $X_{old} = V_0$ (initial velocity)

• Control variables: $m = 200$ (mass)

$\rho = 0.002377$ (density)

$\nabla = 89,300$ (airship volume)

$C_d = 0.044$ (drag coefficient)

L3 integration

L1 state space

Helps determine state elements

• time (t) $x(1)$

• mass (m) $x(4)$

• velocity (V) $x(2)$

• volume (∇) $x(5)$

• density (ρ) $x(3)$

• drag coeff (C_D) $x(6)$

state vector (6x1 matrix)

$\therefore \dot{x} = \text{zeros}(6,1)$

Problem 1.2 (2 pts)

Two Helios spacecraft were deployed in the mid 1970's to study solar processes of the sun. Each spacecraft had a mass of about 370 kg and both were designed to spin at a constant 60 rev/min about the 3-axis. Simulate the spacecraft's initial condition response with 1-axis rotation rate (ω_1) of 0.1 rad/s and the 2-axis rotational rate (ω_2) of 0.01 rad/s. Use the Ralston integration method with step size of 0.05 seconds and run the simulation for a total elapsed time of 30 seconds. Submit an image plot (PNG, BMP, JPG) with 1-axis rotation rate (units: deg/s) on the x-axis and the 2-axis rotational rate (units: deg) on the y-axis ordinate.

$$I_1 = 182 \text{ kg.m}^2 \quad I_2 = 187 \text{ kg.m}^2 \quad I_3 = 194 \text{ kg.m}^2$$

$$I_1 \frac{d\omega_1}{dt} = I_1 \dot{\omega}_1 = (I_2 - I_3) \omega_3 \omega_2$$

$$I_2 \frac{d\omega_2}{dt} = I_2 \dot{\omega}_2 = (I_3 - I_1) \omega_3 \omega_1 \quad I_3 \frac{d\omega_3}{dt} = I_3 \dot{\omega}_3 = 0$$

$$\frac{d\omega_1}{dt} = \left(\frac{I_2 - I_3}{I_1} \right) \omega_3 \omega_2$$

$$\frac{d\omega_2}{dt} = \left(\frac{I_3 - I_1}{I_2} \right) \omega_3 \omega_1$$

$$\frac{d\omega_3}{dt} = 0$$

} EOMs

$$\bullet m = 370 \text{ kg (mass of each spacecraft)}$$

$$\bullet \omega_3 = \frac{60 \text{ rev}}{\text{min}} = \frac{6.2832 \text{ rad}}{\text{s}} \text{ (3-axis rotation rate)}$$

$$\bullet \omega_1 = 0.1 \frac{\text{rad}}{\text{s}} \text{ (1-axis rotation rate)}$$

$$\bullet \omega_2 = 0.01 \frac{\text{rad}}{\text{s}} \text{ (2-axis rotation rate)}$$

$$\text{deg} = \text{rad} \times \frac{180 \text{ deg}}{\pi \text{ rad}}$$

$$\bullet h = 0.05 \text{ (step size)}$$

$$\bullet t_{\text{total}} = 30 \text{ seconds}$$

State Vector

$$\bullet t = x(1)$$

$$\bullet I_1 = x(2)$$

$$\bullet I_2 = x(3)$$

$$\bullet I_3 = x(4)$$

$$\bullet \omega_1 = x(5)$$

$$\bullet \omega_2 = x(6)$$

$$\bullet \omega_3 = x(7)$$

$$\text{Ralston Integration: } x_{k+1} = x_k + \frac{h}{4} (r_1 + 3r_2)$$

$$r_1 = f[x_k] \leftarrow \text{EOM eqns applied to state vector}$$

$$r_1 = \text{EOMS}(x_{\text{old}})$$

$$r_2 = f\left[x_k + \frac{2}{3} h r_1\right]$$

$$r_2 = \text{EOMS}\left(x_{\text{old}} + \frac{2}{3} h r_1\right)$$

Problem 1.3 (2 pts)

L4 Geodesy

Alaska Airlines recently introduced non-stop flights between San Diego CA (KSAN) and Honolulu HI (PHNL). Estimate the total distance between KSAN and PHNL using two different modes of navigation. The **first mode** assumes the airplane travels along a **rhumb line path** between each of the waypoints listed below. The **second navigation mode** assumes the pilot ignores the intermediate waypoints and follows a **great circle path** between KSAN and PHNL. Report as your answer the magnitude of the difference between the total trip distance computed by each mode of navigation. **Ignore aircraft altitude** and assume a spherical Earth with **semi-major axis equal to 6378137 m**.

Name	Latitude (deg)	Longitude (deg)	
KSAN	32.7335556	-117.1896667	1
ETECO	30.2858333	-131.6361111	2
DIALO	28.7016667	-139.4266667	3
DUSAC	26.0061111	-146.1675000	4
DRAYK	23.0241667	-152.5777778	5
PHNL	21.3178275	-157.9202627	6

deg → rad
 $\left[\text{rad} = \frac{\pi}{180} \text{deg} \right]$ conversion factor
 for great circle method only
 for rhumb line method only
 $|d_{gc} - d_{rh}|$
 magnitude of trip distance difference = 10.0847 [km]

$a = 6378137 \text{ m}$ (semi-major axis)

Great circle path: $d_{gc} = a \cos^{-1} [\sin \varphi_A \sin \varphi_B + \cos \varphi_A \cos \varphi_B \cos (\lambda_B - \lambda_A)]$ * no iterations needed

(φ_A, λ_A) : (lat, long) of start point ← very first point (KSAN)

(φ_B, λ_B) : (lat, long) of end point ← very last point (PHNL)

Rhumb line path: $d_{rh} = a |\varphi_B - \varphi_A| \sec \varphi$ where $\varphi = \tan^{-1} \left(\frac{\lambda_B - \lambda_A}{\tau_B - \tau_A} \right)$ * process repeated for each segment (5 segments total here)

(φ_A, λ_A) : (lat, long) of segment start point

(φ_B, λ_B) : (lat, long) of segment end point

$$\tau_A = \ln (\sec \varphi_A + \tan \varphi_A)$$

$$\tau_B = \ln (\sec \varphi_B + \tan \varphi_B)$$

Problem 1.4 (2 pts) L4 Geodesy, LS Velocity

A private pilot plans to fly a rhumb line course from San Diego airport (KSAN) to San Francisco airport (KSFO). If there is no wind, the pilot expects the trip to take about 130 minutes at a airspeed of 180 nmi/hr. However, when the actual flight occurs, the wind is blowing (from west) toward east at 20 nmi/hr. How long will the flight actually take if the pilot maintains the 180 nmi/hr airspeed? (1 nmi = 6076 ft)

Name	Latitude (deg)	Longitude (deg)
KSAN	32.7336	-117.1897
KSFO	37.6167	-122.3750

- a. 120 minutes c. 140 minutes
b. 130 minutes d. 160 minutes

For True Course (χ_G):

Rhumb line path: $\chi_G = a |\varphi_B - \varphi_A| \sec \varphi$ where $\varphi = \tan^{-1} \left(\frac{\lambda_B - \lambda_A}{\tau_B - \tau_A} \right)$

(φ_A, λ_A) : (lat, long) of segment start point

(φ_B, λ_B) : (lat, long) of segment end point

$$\chi - \chi_G = \sin^{-1} \left[\frac{W_N}{V} \sin(\chi_G) - \frac{W_E}{V} \cos(\chi_G) \right]$$

$W_N = 0$ (no north wind)

$$\chi - \chi_G = \sin^{-1} \left[-\frac{W_E}{V} \cos(\chi_G) \right]$$

V : airspeed provided
 W_E : windspeed provided

sub in

$$V_G = V \cos(\chi - \chi_G) + W_E \sin(\chi_G) + \cancel{W_N \cos(\chi_G)}$$

$W_N = 0$ (no north wind)

$$V_G = V \cos(\chi - \chi_G) + W_E \sin(\chi_G)$$

Flight Duration: $t = \frac{\chi_G}{V_G} = \frac{\chi_G}{V \cos(\chi - \chi_G) + W_E \sin(\chi_G)}$

* Convert [m] $\rightarrow$ [nmi]
[nmi/hr] $\rightarrow$ [nmi/min]

$$1852m = 1 \text{ nmi}$$

deg $\rightarrow$ rad

$$\text{rad} = \frac{\pi}{180} \text{ deg}$$

$$a = 6378137m \text{ (assumption)}$$

* process repeated for each segment (1 segment total here)

$$\tau_A = \ln(\sec \varphi_A + \tan \varphi_A)$$

$$\tau_B = \ln(\sec \varphi_B + \tan \varphi_B)$$

Problem 1.5 (2 pts)

Estimate the ratio of atmospheric density at Coors Field in Denver CO (altitude = 5280 ft), on a non-standard "hot" day, relative to the sea-level or base level for the same day.

↑ use non-standard atmosphere

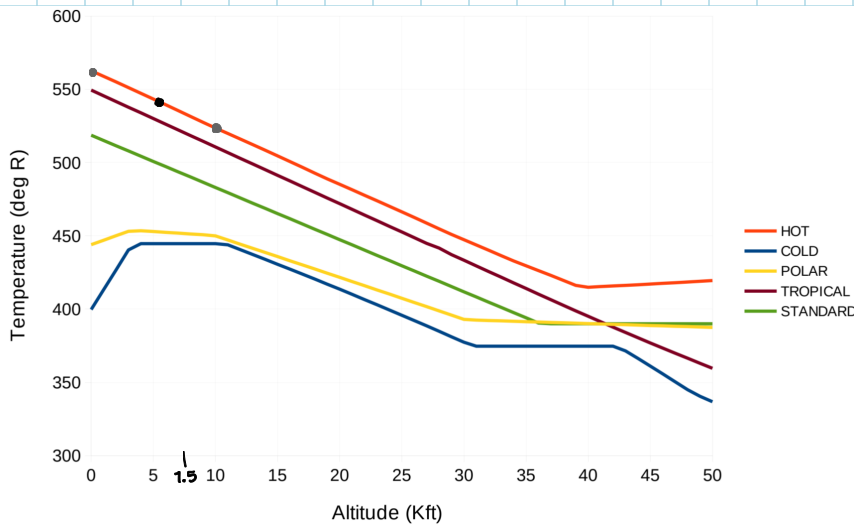
$$\text{density ratio} = \rho/\rho_0 = \rho(5280)/\rho(0)$$

- a. 0.83 c. 1.17
 b. 0.86 d. 1.20

$$\frac{\rho}{\rho_0} = \left[\frac{T(5280\text{ft})}{T(\text{base})} \right]^{-\left[1 + \frac{g}{aR} \right]}$$

where $g = 32.2 \text{ ft/s}^2$
 $R = 1716 \frac{\text{ft} \cdot \text{lb}}{\text{slug} \cdot ^\circ\text{R}}$

$$a = \frac{dT}{dh} = \text{lapse rate (for "hot day" in this problem)}$$



(0 ft, 560°R)

(10,000 ft, 525°R)

$$\frac{525^\circ\text{R} - 560^\circ\text{R}}{10,000\text{ft} - 0\text{ft}} = \underbrace{-\frac{7}{2000}}_a \frac{^\circ\text{R}}{\text{ft}}$$

$$\cdot T(5280\text{ft}) \sim 540^\circ\text{R}$$

$$\cdot T(\text{base}) \sim 560^\circ\text{R}$$